

MTH501 FINAL TERM BY IQRA AZHAR

MTH501

Final Term

2024

Subjective & Mcqs

SOLVED BY

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YTUBE Mth zone for vu

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If two matrixes are added, then which of followings should be true for them? Both has same order

If $AB = I = BA$ for matrices A where, B and I, I is an identity matrix, then

B is Inverse of A A is inverse of B $A^{(-1)} = B, B^{(-1)} = A$
all

Let A be the matrix of order 2×3 and B be the matrix order 3×5 , then which of the following is the order of the matrix $A * B = ?$ 2×5

The dominate for the matrix $A = \begin{bmatrix} 0 & 0 & 2 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ -3

Which statement about the matrix $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 7 & 2 & 0 & 0 \\ 9 & 1 & 2 & 0 \\ 5 & 4 & 2 & -1 \end{bmatrix}$ is false Eigen value 2 has algebraic multiplicity 1

multiplicity 1

A square matrix A is said to be diagonal if A is similar to a matrix Diagonal matrix

An $n \times n$ matrix A is said to be diagonalizable if and only if A has n linearly independent eigenvectors.

A 3×3 identity matrix has three and ----- eigenvalues. Same

Consider the bases for $\mathbb{R}^3$ given by $B = ({}^r b_1, {}^r b_2)$ or $C = ({}^r C_1, {}^r C_2)$ where

$${}^r b_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, {}^r b_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}, {}^r C_1 = \begin{bmatrix} -5 \\ 3 \end{bmatrix}, {}^r C_2 = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

also assume that $P_{B \leftarrow C} = \begin{bmatrix} -2 & -1 \\ 3 & 1 \end{bmatrix}$ then which of the following is the change of coordinates matrix from B to C?

$$P_{B \leftarrow C} = \begin{bmatrix} -5 & -2 \\ 3 & 1 \end{bmatrix}$$

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if the matrix $A = \begin{bmatrix} 1 & 5 & 4 \\ 0 & 1 & 7 \\ 0 & 0 & 0 \end{bmatrix}$ then which of the Its determinant is 0

If $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \\ 0 & -1 & 1 \end{bmatrix}$ then which of the following is true for the matrix A? It is non invertible matrix

if a matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 6 & 1 & 1 \end{bmatrix}$ which of the following is true for A ? it is singular matrix

if A is an $m \times n$ matrix with linearly independent column vector, then A can be factored as $A = QRQ$ is an $m \times n$ matrix orthonormal column vector and R is an $m \times m$. Invertible Upper triangular matrix

Let $v = (1, -2, 2, 0)$. The unit vector in the same direction as v is: $\frac{1}{3}, \frac{-2}{3}, \frac{2}{3}, 0$

Let $v = (0, 2, 2, 1)$. The unit vector in the same direction as v is: $0, \frac{1}{3}, \frac{2}{3}, \frac{2}{3}$

Let I^3 have the Euclidean inner product. Then $u = (2, 1, 3)$, $V = (1, 7, k)$ are ... orthogonal for $K = -3$

Let $u = (3, -2)$, $v = (4, 5)$. For the weighted Euclidean inner product $(u, v) = \dots 4u_1v_1 + 5u_2v_2 = -2$

Let A be an $n \times n$ Matrix whose entries are real. If λ is an eigenvalue of A with corresponding eigenvector in $\mathbb{R}^n$, then $\overline{Ax} = \overline{\lambda x}$

Suppose that $A = \begin{bmatrix} 1.25 & -.75 \\ -.75 & 1.25 \end{bmatrix}$ has eigenvalues 2 and 0.5. then origin is Saddle point

A column replacement operation on A does not change the Saddle

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Suppose that $A = \begin{bmatrix} 0.5 & 0.6 \\ -0.3 & 1.4 \end{bmatrix}$ has eigenvalues 0.8 and then Origin is a Saddle point

The matrix equation $A^T A \hat{x} = A^T b$ represent a system of linear equation
Commonly referred to as the Normal equation for $\hat{x}$

By the best Approximation theorem, the distance from y to W is $\|y - Y^\wedge\|$,
where $Y^\wedge = \text{Proj}_W Y$

$\|u + v + w\| \leq \|u\| + \|v\| + \|w\|$ for all vectors u, v and w in an inner product space True

A square matrix A is invertible if and only if $\det A \neq 0$ is not an Eigen
Value of A True

A square matrix with orthogonal columns -----matrix is orthogonal

if X is orthogonal to both U and V , then must be -----to $u+v$ orthogonal

if two rows are orthogonal, are..... they Linearly independent

Let ' $Ax = 0$ ' be a homogeneous linear system of ' n ' equations and ' n ' unknowns. Then, the
coefficient matrix ' A ' is invertible if and only if this system has ---solution. Trivial

the given system $\begin{cases} 2x + 3y = 3 \\ 6x + 9y = 7 \end{cases}$ has no solution

if $A = \begin{bmatrix} \partial_{11} & \partial_{12} \\ \partial_{21} & \partial_{22} \end{bmatrix}$ is diagonalizable A has 2 distinct eigenvalues true

A is diagonalizable if $A = PDP^{-1}$ where D is diagonalizable and P is invertible.

How many terms are there in the algebraic expression $8x^2 + \sqrt{9x} \times 25x^3$ 2

if v^1, v^2, v^3 are in $\mathbb{R}^m$ then which of the following is equivalent to $\begin{bmatrix} v_1 & v_2 & v_3 \end{bmatrix} \begin{bmatrix} 2 \\ -7 \\ 5 \end{bmatrix} = ?$

$$2v^1 - 7v^2 + 5v^3$$

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If $(^r v_1, ^r v_2, ^r v_3)$ is linearly dependent set and $^i v_1 \neq C^i v_2$ (where 'c' is a scalar), which option is true?

$$^r v_3 \in \text{span} (^r v_1, ^r v_2)$$

Which statement about the General Least Square Method is true?

This method find an X that make Ax as close possible to the b

which of the following is true for the partitioned matrices $A = \begin{pmatrix} C & D \end{pmatrix}$ and $B = \begin{pmatrix} E & F \end{pmatrix}$, where Sub-matrices C and D have the same size As E and F respectively? $A+B = \begin{bmatrix} C+E & D+F \end{bmatrix}$

If a matrix A is factorized into lower and upper triangular matrices, then which of the following is true for T The matrix? It is called an LU- Decomposition

Let a set S is a basis of a vector space V, then which of the following is NOT true about it It is linearly independent

If $B = \left\{ \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 0 \end{pmatrix} \right\}$ for R^2 and $x^1 \in R^2$ has coordinate vector $[x^r]_B = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$ which of the following is the value of x ?

$$x^r = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

if a $s = \{1, x, x^2\}$ is a for p_2 and $[p^1]_s = (2, 4, 7)$, then which of the following is the most appropriate option? $P_2 = 2 + 4x + 7x^2$

which of the following is the set of standard basis for R^3 ? $\{(1, 0, 0), (0, 1, 1), (0, 0, 1)\}$

If the general term of a typical signal is $(0.6)^k$, then determine which of the following is the signal for $k = -2$? $(0.6)^{-2} = 1/(0.6)^2$

if the casorati matrix is not is not invertible, then which of the following is the most appropriate option regarding the associated signals The signals are linearly Independent.

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if $\{Y_k\} = \{\dots, 1, 0.7, 0, -0.7, 0, 0.7, 1, 0.7, 0, \dots\}$ and $0.35Y_{k+2} + 0.6Y_{k+1} + 0.42Y_k = Z_k$; $K=0$

Then which of the following is the value of Z_0 ? **0.840**

A system of linear equation is said to be homogenous if it can be written in the form---
----- **$AX = 0$**

If $X - 2$ is a factor of the characteristic polynomial of matrix C then an eigenvalue of C is. **2**

$\lambda + 2$ is a factor of the characteristic polynomial of matrix C . then which of the following is the eigenvalue of C ? **-2**

Let A and B be the square matrices. Then A and B are invertible with $B = A^{-1}$ and $A = B^{-1}$ if and only if $AB = BA$ equals to a (an) ----- matrix. **Identity**

Let V be a five dimensional vector space let S be a subset of V which spans V Then **Must have 5 elements**

If $U + V = U + W$ then **$V = W$**

What is Eigen value? **A scalar associated with given linear transformation.**

Let a matrix A has both negative and positive eigenvalues, so in this case origin behaves as a.....point **Saddle**

A matrix $A - (n \times n)$ has both positive and negative eigenvalues so in this case origin behave as **Saddle point**

a matrix A has both negative and positive Eigenvalues so in this case origin behaves **point Saddle**

vectors u and v in R^n are orthogonal if..... **$u \cdot v = 0$**

A vector whose length 1 is called **Unit vector**

A Matrix with orthogonal columns is an orthogonal matrix. **Square 1:**
if a square matrix has orthogonal columns then it also has rows. **Orthogonal**

The norm of v is the non-negative scalar $\|v\|$ defined by

$$\|v\| = \sqrt{v_1^2 + v_2^2 + \dots + v_n^2}$$

the matrix $A^T \times A$ is invertible if and only if the columns of A are linearly independent **TRUE**
let V be a one Eigen vector then conjugate eigenvector is represented by. **V**

the vector is orthogonal to every vector in R^n . **Zero**

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Each pair of eigenvalue and this its corresponding eigenvector provides a solution of the equation $X' = Ax$

which is called of the differential equation. **Eigen function**

if we divide a non – zero vector by its length we get a **Unit vector**

Let A be a real 2 by 2 matrix with complex eigen values $\lambda = a - b + i(b \neq 0)$ and associated eigenvectors v in C^2 then $AP = P^{-1}$ where $P = [Re\ v\ Im\ v]$

Let $Ax = 0$ be a homogeneous linear system of 'n' unknown, then the coefficient matrix 'A' is invertible if and only if this system has.....solution **Trivial**

The two vector are said to be equivalent if **Same length and same direction**

if one of the eigenvalues of $[A]_{n \times n}$ is zero .it implies **The determinant of [A] is zero**

let be a five – dimensional vector space. And let be a subset of consisting of five vectors.

Must be linearly dependent, but may or may not span V

let U, V and W be vectors in R^n , then **$(U + v) \cdot W = U \cdot W + V \cdot W$**

if u and v non zero vector in either R^2 or R^3 then by the law of cosines $\|u - v\|^2 = \|u\|^2 + \|v\|^2 - 2 \|u\| \|v\| \cos$

If $u + v = w$, then **$V = w$**

Let W be a subspace of R^n and $(u_1, u_2, \dots, u_p)$ is any orthogonal basis of W, then $y = c_1 u_1 + \dots + c_p u_p$

where **$C_j =$**

Two vector are if at least one of the vector is a multiple of the other **Linearly dependent**

Electric circuit rotation is caused by sine and cosine function if there eigenvalues are complex and hence the origin is called.... **Spiral point**

Multiplication of a partitioned matrix by a scalar is also computed.... **Column by column**

The invertible Matrix Theorem applies only to..... **Square**

If a system of equations is solved using the Jacobi's method, then which of the following is the most appropriate answer about the matrix M that is derived from the coefficient matrix ?

All **of its entries below and above the diagonal must be zero.**

How many different permutations are there in the set of integers $\{1, 2, 3\}$? **6**

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Which one is the numerical method used for approximation of dominant eigenvalue of a matrix **Gauss Sedial Method**

The inverse of an invertible lower triangular matrix is. **Lower triangular matrix**

If the augmented matrices of two linear systems are row equivalent, then the two systems have the same solution set. **true**

Every linear transformation is a matrix transformation. **True**
Suppose x, y, z are some vectors in an inner product space $(V, \langle \cdot, \cdot \rangle)$ such that $\langle X, Y \rangle = \langle X, Z \rangle$ for all $x \in V$ then $Y=Z$ **TRUE**
Any finite dimensional inner product space has an orthonormal basis. **True**

A null space is a vector space **TRUE**

Two equivalent vectors must have the same initial point. **False**

All the lines that pass through origin are not the subspace of a plane.

Why inverse of the matrix $A = \begin{bmatrix} 1 & 2 \end{bmatrix}$ is NOT possible? **Because it is a rectangle matrix**

If λ is an eigenvalue of A , then every nonzero vector x such that $Ax = \lambda x$ is called an ----- of A corresponding to --- **Eigenvector. λ**

Let A be an $n \times n$ matrix, then A is invertible if and only if **$\det A \neq 0$**

The invertible matrix theorem applies only to ----- matrices. **SQUARE**
point

Each pair of eigenvalue and its corresponding eigenvector provides a solution of the equation $X' = AX$ which is called **Eigen FUNCTION**

If A is invertible and $b \in \mathbb{R}^n$ be any vector. Then, we must have a matrix A^{-1} b , which is a solution of --- **$AX=b$**

What is the maximum possible number of pivots in a 6×6 matrix? **6**

If 3 is an eigenvalue of A and x is the corresponding eigenvector, then what is the eigenvalue of A^2 ?

The characteristic polynomial of a 3×3 identity matrix is ----- if x is the eigenvalue of the given **3×3**
identity matrix. **$(x-1)^3$**

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A partitioned matrix 'A' is said to be block diagonal if the matrices on the main diagonal are square and all other position matrices are 0

If A is an invertible square matrix then $(A^T)^{-1} = (A^{-1})^T$

A blocked matrix in which blocks are repeated down the diagonals of the matrix is called a block-diagonal matrix.

Blocked diagonal constant

The characteristic polynomial of a 5*5 matrix is, $\lambda^5 - 4\lambda^4 - 45\lambda^3 = 0$ 0,0,0,-5,9

A partitioned square matrix 'A' is said to be the matrix if the matrices on the main diagonal are square and all matrices above the main diagonal are zero. Block lower triangular

If there is a vector $v = (2, 1, 0)$ then $\|v\|$ isSQR..5

λ is an eigenvalue of a matrix A if and only if the equation $(A - \lambda I)x = 0$ has a non-trivial solution

Which of the following is the eigenvalue of the matrix? The complex conjugate of a vector in C^n is the vector x in C^n

Whose entries are the conjugates of the entries in x complex

Multiplication of a partitioned matrix by a scalar is also computed block by block

If the real part of the eigenvalue is zero, then the trajectories from around the origin. Ellipse

A row interchange changes the sign of the determinant. Change

If both the Jacobi and Gauss-Seidel sequences converge for the solution of $Ax=b$, for any initial $x(0)$, then which

of the following is true about both the solutions? unique solution

We can add the matrices of the same order

Solving a system of equations with an iterative method, we stop the process when the entries in two

successive iterations are the same repeat

Jacobi's Method is slower than Gauss-Seidel Method slow

Let T be any $m \times n$ matrix with orthonormal columns and v be any vector then $\|Tv\| = \|v\|$.

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Each pair of eigenvalue and its corresponding eigenvectors provide solution of equation. **Eigenfunction**

which of the following is true for the matrix Where M_{11} , M_{22} and M_{33} are square submatrices, and O is Zero sub matrix? **It is block upper triangular matrix**

The determinant of A is the product of the pivots in any echelon form U of A , multiplied by $(-1)^r$, Where r is **the number of rows of U**

suppose that real solution y_1 and y_2 of $x' = Ax$ form a basis for the two-dimensional real vector space if y_1 and y_2 are **Linearly independent**

Elementary row operation on a matrix do not affect the relation among the column of the matrix

Linear dependence

Subjective

Question No : 41 of 52

Determine whether the set of vectors are orthogonal or not.

$$\begin{bmatrix} 5 \\ -4 \\ 0 \\ 3 \end{bmatrix}, \begin{bmatrix} -4 \\ 1 \\ -3 \\ 8 \end{bmatrix}, \begin{bmatrix} 3 \\ 3 \\ 5 \\ 1 \end{bmatrix}$$

$$\text{let } u = \begin{bmatrix} 5 \\ -4 \\ 0 \\ 3 \end{bmatrix}, v = \begin{bmatrix} -4 \\ 1 \\ -3 \\ 8 \end{bmatrix}, w = \begin{bmatrix} 3 \\ 3 \\ 5 \\ 1 \end{bmatrix}$$

$$u.v = \begin{bmatrix} 5 \\ -4 \\ 0 \\ 3 \end{bmatrix} \cdot \begin{bmatrix} -4 \\ 1 \\ -3 \\ 8 \end{bmatrix}$$

$$u.v = -20 - 4 - 0 + 24 =$$

$$u.v = 0 \quad \quad \quad u.v \text{ is orthogonal}$$

$$v.w = \begin{bmatrix} -4 \\ 1 \\ -3 \\ 8 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 3 \\ 5 \\ 1 \end{bmatrix}$$

$$v.w = -12 + 3 - 15 + 8$$

$$v.w = -15 \quad \quad \quad \text{is not orthogonal}$$

$$u.w = \begin{bmatrix} 5 \\ -4 \\ 0 \\ 3 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 3 \\ 5 \\ 1 \end{bmatrix}$$

$$u.w = 15 - 12 + 0 + 3$$

$$u.w = 6 \quad \quad \quad \text{is not orthogonal}$$

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Is the following set of vectors is orthogonal with respect to the Euclidean inner product on $\mathbb{R}^3$

$$\left\{ \left(\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, -\frac{2}{\sqrt{6}} \right), \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0 \right) \right\}$$

$$\left(\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, -\frac{2}{\sqrt{6}} \right) \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0 \right)$$

$$u.v = u_1v_1 + u_2v_2 + u_3v_3$$

$$u.v = \left[\frac{1}{\sqrt{6}} \cdot \frac{1}{\sqrt{2}} \right] + \left[\frac{1}{\sqrt{6}} \cdot -\frac{1}{\sqrt{2}} \right] + \left[\frac{-2}{\sqrt{6}} \cdot 0 \right]$$

$$u.v = \frac{1}{\sqrt{12}} - \frac{1}{\sqrt{12}} - 0$$

$$u.v = 0 \text{ it is orthogonal}$$

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Find the dominant Eigen pair (i.e. the Eigen value and Eigen vector) by using the Power Method for the following matrix

$$A = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}, \quad x_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

1 Iteration

$$A = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix} \quad x_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$Ax_0 = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$Ax_0 = \begin{bmatrix} 5 + 0 \\ 1 + 0 \end{bmatrix}$$

$$Ax_0 = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$$

$$\|Ax_0\| = \sqrt{5^2 + 1} \Rightarrow \|Ax_0\| = \sqrt{25 + 1}$$

$$\|Ax_0\| = \sqrt{26}$$

$$\frac{Ax_0}{\|Ax_0\|} = \frac{\begin{bmatrix} 5 \\ 1 \end{bmatrix}}{\sqrt{26}} \Rightarrow \begin{bmatrix} 5 / \sqrt{26} \\ 1 / \sqrt{26} \end{bmatrix}$$

$$= \frac{1}{\sqrt{26}} \begin{bmatrix} 5 \\ 1 \end{bmatrix}$$

$$\text{eigen vector is } = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$$

$$\text{eigen value } = \lambda = \sqrt{26} = 5.099$$

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If a matrix $A = \begin{bmatrix} 1 & 4 & 5 \\ 4 & 5 & 6 \\ 7 & 8 & 8 \end{bmatrix}$ and $\det(A^t) = 6$, then find the determinant of the matrix.

$$A = \begin{bmatrix} 1 & 4 & 5 \\ 4 & 5 & 6 \\ 7 & 8 & 8 \end{bmatrix}$$

$$\det A = 1 \begin{bmatrix} 5 & 6 \\ 8 & 8 \end{bmatrix} - 4 \begin{bmatrix} 4 & 6 \\ 7 & 8 \end{bmatrix} + 5 \begin{bmatrix} 4 & 5 \\ 7 & 8 \end{bmatrix}$$

$$\det A = 1(40 - 48) - 4(32 - 42) + 5(32 - 35)$$

$$\det A = 1(-8) - 4(-10) + 5(-3)$$

$$\det A = -8 + 40 - 15$$

$$\det A = 17$$

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Let $B = \{b_1, b_2, b_3\}$ and $D = \{d_1, d_2\}$ be bases for vector spaces V and W , respectively. Let $T: V \rightarrow W$ be a linear transformation such that $T(b_1) = 3d_1 - 5d_2$ and $T(b_2) = -d_1 + 6d_2$ and $T(b_3) = 4d_2$. Find a matrix M for T relative to B and D .

Marks: 2 (Budgeted Time 4 Min)

ctively. Let $T: V \rightarrow W$ be a linear transformation with the property that $T(b_1) = 3d_1 - 5d_2$,

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Determine whether the vectors $y = \begin{bmatrix} -3 \\ 7 \\ 4 \\ 0 \end{bmatrix}$, $z = \begin{bmatrix} 1 \\ -8 \\ 15 \\ -7 \end{bmatrix}$ are orthogonal.

Answer ([Please click here to Add Answer](#))

$$Y = \begin{bmatrix} -3 \\ 7 \\ 4 \\ 0 \end{bmatrix} \quad Z = \begin{bmatrix} 1 \\ -8 \\ 15 \\ -7 \end{bmatrix}$$

$$Y.Z = \begin{bmatrix} -3 \\ 7 \\ 4 \\ 0 \end{bmatrix} \begin{bmatrix} 1 \\ -8 \\ 15 \\ -7 \end{bmatrix}$$

$$Y.Z = -3 - 56 + 60 - 0$$

$$Y.Z = 1 \quad \text{NOT ORTHOGONAL}$$

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If $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$ then find an invertible matrix P such that $P^{-1}AP = D(\text{diagonal matrix})$

$$A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$$

$$\det(A - \lambda I) = 0 \dots 1$$

$$A - \lambda I = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$A - \lambda I = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}$$

$$A - \lambda I = \begin{pmatrix} 1 - \lambda & 2 \\ 2 & 1 - \lambda \end{pmatrix}$$

$$\det(A - \lambda I) = (1 - \lambda)^2 - 4$$

$$\det(A - \lambda I) = (1 - 2\lambda + \lambda^2) - 4$$

$$\det(A - \lambda I) = \lambda^2 - 2\lambda - 3$$

$$\lambda^2 - 2\lambda - 3$$

$$\lambda^2 - 3\lambda + 1\lambda - 3$$

$$\lambda(\lambda - 3) + 1(\lambda - 3)$$

$$(\lambda - 3)(\lambda + 1)$$

$$\lambda + 1 = 0$$

$$\lambda - 3 = 0$$

$$\lambda = -1$$

$$\lambda = 3$$

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when $\lambda = 3$

$$(A - \lambda I)x = 0$$

$$\begin{pmatrix} 1-\lambda & 2 \\ 2 & 1-\lambda \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 1-3 & 2 \\ 2 & 1-3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} -2 & 2 \\ 2 & -2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$-2x_1 + 2x_2 = 0$$

$$x_1 = x_2$$

So, the eigenvector corresponding to $x_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

when $\lambda = -1$

$$(A - \lambda I)x = 0$$

$$\begin{pmatrix} 1-\lambda & 2 \\ 2 & 1-\lambda \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 1+1 & 2 \\ 2 & 1+1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$2x_1 + 2x_2 = 0$$

$$2x_1 + 2x_2 = 0$$

$$x_1 = -x_2$$

So, the eigenvector corresponding to $\lambda = -1$ is $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$

$$P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

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Let $W = \text{Span} \{x_1, x_2\}$, where $x_1 = \begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix}$, $x_2 = \begin{bmatrix} 8 \\ 5 \\ -6 \end{bmatrix}$. Construct an orthogonal basis $\{v_1, v_2\}$ for W .

$$v_1 = x_1 = \begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix}$$

$$x_2 = \begin{bmatrix} 8 \\ 5 \\ -6 \end{bmatrix}$$

$$v_2 = x_2 - P$$

$$P = \frac{x_2 v_1}{v_1 v_1}$$

$$P = \frac{\begin{bmatrix} 8 \\ 5 \\ -6 \end{bmatrix} \begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix}}{\begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix} \begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix}} \begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix}$$

$$P = \frac{24+0+6}{9+0+1} \begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix} \Rightarrow P = \frac{30}{10} \begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix}$$

$$P = \begin{bmatrix} 9 \\ 0 \\ -3 \end{bmatrix}$$

$$v_2 = x_2 - P = \begin{bmatrix} 8 \\ 5 \\ -6 \end{bmatrix} - \begin{bmatrix} 9 \\ 0 \\ -3 \end{bmatrix}$$

$$v_2 = \begin{bmatrix} -1 \\ 5 \\ -3 \end{bmatrix}$$

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Compute the least square error associated with the least square solution $\hat{x} = \begin{bmatrix} \frac{4}{3} \\ -\frac{1}{3} \end{bmatrix}$ of the equation $Ax = b$ where $A = \begin{bmatrix} 1 & -2 \\ -1 & 2 \\ 0 & 3 \\ 2 & 5 \end{bmatrix}$, $b = \begin{bmatrix} 3 \\ 1 \\ -4 \\ 2 \end{bmatrix}$

$$b - Ax$$

$$Ax = \begin{bmatrix} 1 & -2 \\ -1 & 2 \\ 0 & 3 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 4/3 \\ -1/3 \end{bmatrix}$$

$$Ax = \begin{bmatrix} 1 \times 4/3 & -2 \times -1/3 \\ -1 \times 4/3 & + 2 \times -1/3 \\ 0 \times 4/3 & + 3 \times -1/3 \\ 2 \times 4/3 & + 5 \times -1/3 \end{bmatrix}$$

$$Ax = \begin{bmatrix} 4/3 + 2/3 \\ -4/3 - 2/3 \\ 0/3 - 3/3 \\ 8/3 - 5/3 \end{bmatrix} \Rightarrow Ax = \begin{bmatrix} 6/3 \\ -6/3 \\ -3/3 \\ 3/3 \end{bmatrix} \Rightarrow Ax = \begin{bmatrix} 2 \\ -2 \\ -1 \\ 1 \end{bmatrix}$$

$$b - Ax = \begin{bmatrix} 3 \\ 1 \\ -4 \\ 2 \end{bmatrix} - \begin{bmatrix} 2 \\ -2 \\ -1 \\ 1 \end{bmatrix} \Rightarrow b - Ax = \begin{bmatrix} 1 \\ 2 \\ -3 \\ 1 \end{bmatrix}$$

$$\|b - Ax\| = \sqrt{(1)^2 + (2)^2 + (-3)^2 + (1)^2}$$

$$\|b - Ax\| = \sqrt{1 + 4 + 9 + 1}$$

$$\|b - Ax\| = \sqrt{15}$$

$$\|b - Ax\| = 3.87299$$

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Find A^2 , given that $A = PDP^{-1}$, where P and D are given below $A = \begin{pmatrix} 2 & 6 \\ -4 & 12 \end{pmatrix}$, $P = \begin{pmatrix} 3 & 1 \\ 2 & 1 \end{pmatrix}$, $D =$

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$$), D = \begin{pmatrix} 6 & 0 \\ 0 & 8 \end{pmatrix}, P^{-1} = \begin{pmatrix} 1 & -1 \\ -2 & 3 \end{pmatrix}$$

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Let $A = \begin{bmatrix} 1 & 4 \\ 5 & 6 \end{bmatrix}$ and $\vec{c} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$. Define $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by $T(\vec{x}) = A\vec{x}$. Determine if $\vec{c}$ is in the range of the transformation T .

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is in the range of the transformation T .

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